

Part 2:- Feature Selection | Wrapper methods

Wrapper methods are feature selection techniques that use a machine learning model to decide which features are important.

→ Instead of looking at features individually

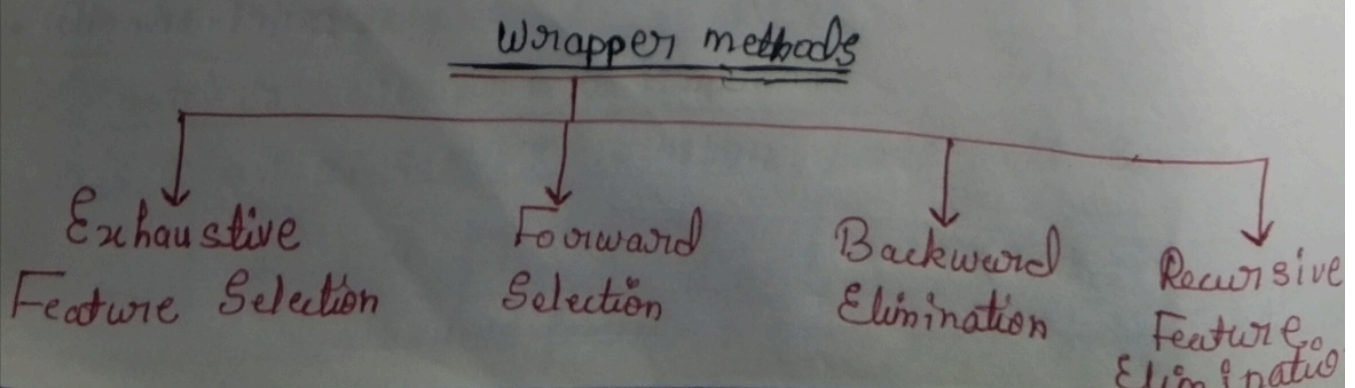
→ They Evaluate feature subset using model performance

[Wrapper methods select features by training a model and checking how well it performs with different feature combinations]

Feature subset → Train model → Check Performance → Decide

How Wrapper Methods Work:-

- i) Choose ML model
- ii) Select subset of features → start with No features OR all features.
- iii) Train the model → Train the chosen model using only those features.
- iv) Evaluate model Performance → Performance tells how good the selected features are
- v) Modify Feature set → add a feature, Remove a feature or replace a feature
- vi) Repeat the Process:-
 - Try different feature combination
 - Keep the combination that gives best performance
 - A feature is good, if the model performance Improves when it is included.

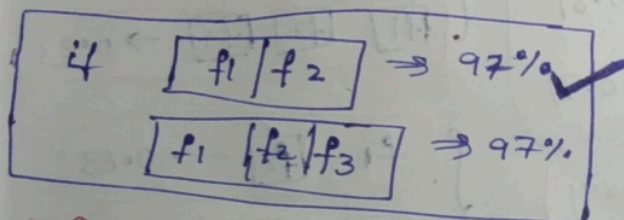
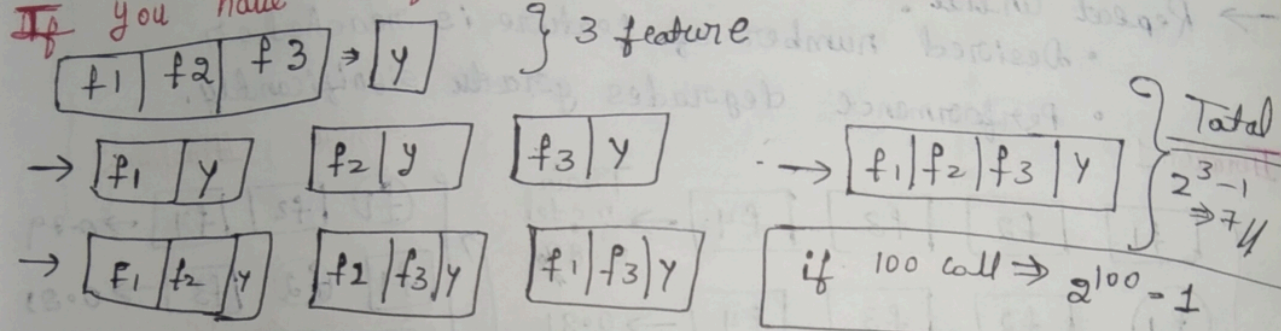


Exhaustive Feature Selection / Best subset Selection :-

Exhaustive Feature Selection is a wrapper method that tries all possible combination of features and selects the subset that gives the best model performance.

checks every possible feature subset $\rightarrow$ train model $\rightarrow$ pick the best one

If you have n features, it evaluates $2^n - 1$ subsets



Since we are doing feature selection, we should select the min features

$\rightarrow$ Partially impossible for large datasets.

Disadvantages :-

- $\rightarrow$ Extremely Computationally Expensive
- $\rightarrow$ Not Scalable (features $\leq 15, 20$)
- $\rightarrow$ Very Time Consuming
- $\rightarrow$ High memory and resource usage
- $\rightarrow$ Risk of overfitting :- Problematic without proper cross validation.

advantage

- will not miss any feature combo
- Best for small dataset
- gives best accuracy for small data

Sequential Backward Selection / Elimination :-

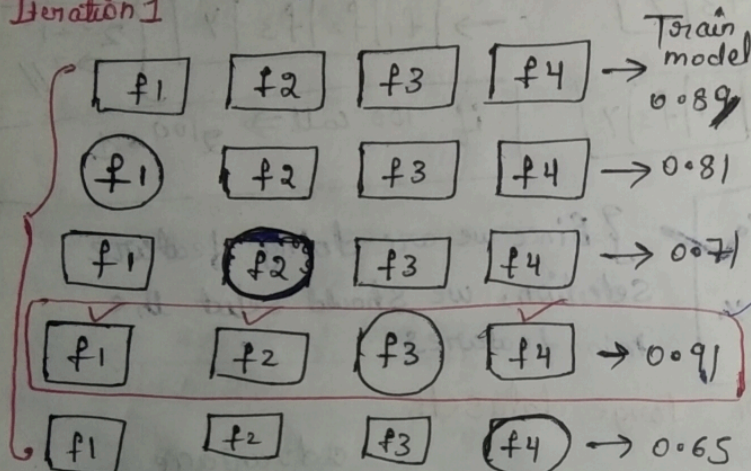
Sequential Backward Selection is a wrapper feature selection method that starts with all features and removes them one by one based on model performance.

start with every thing $\rightarrow$ remove the least useful feature step by step.

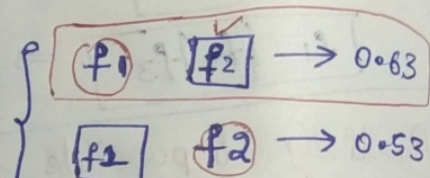
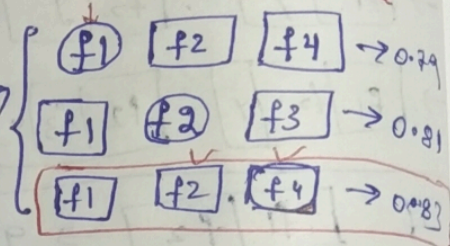
HOW it works :-

- Start with all features
- Train a model
- Remove one feature at a time
- Evaluate model performance after each removal
- Permanently remove the feature whose removal improves or least hurts performance
- Repeat until:
 - Desired number of feature is reached or
 - Performance degrades gradually significantly.

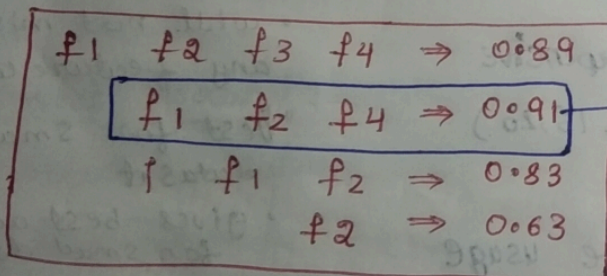
Iteration 1



Iteration 2



Iteration 3



best feature subset

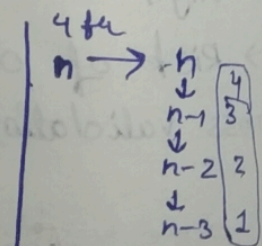
Train model $\Rightarrow$ result

• not trying all combos
advantage:- faster than Exhaustive ($2^n - 1$)

feature	model training
4 $\rightarrow$ 4	1 $\rightarrow$ 1
3 $\rightarrow$ 3	
2 $\rightarrow$ 2	

$\Rightarrow 1 + 2 + 3 + \dots + n$

$\Rightarrow \frac{n(n+1)}{2}$



→ for 100 cols:- $(2^{100} - 1)$ $\frac{100 \times 101}{2} \Rightarrow 5051$

Disadvantage:-

Cannot recover a removed feature

Slower than forward selection

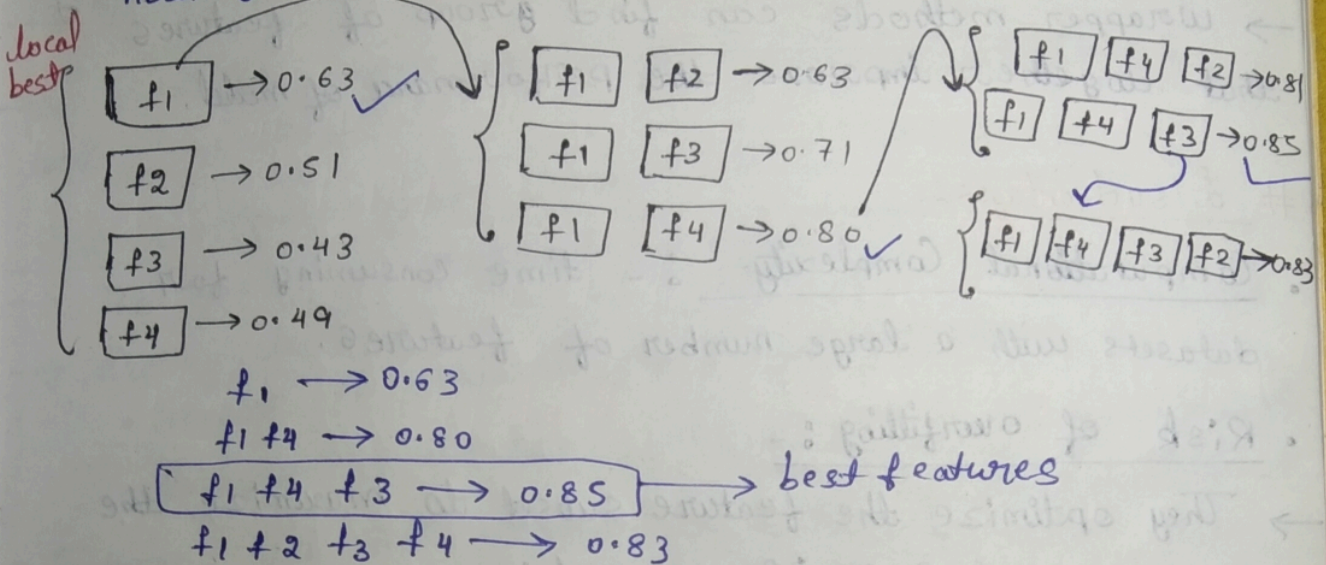
Computational Expensive

may miss local feature (f3), $\Rightarrow$ missing best chance of local

Sequential Forward Selection:-

Forward Selection starts with no features and adds features one by one based on model performance.

starts with nothing → add the most useful feature step by step.



n-features:-

$$\left\{ \begin{array}{ll} \text{Exhaustive} & \rightarrow 2^n - 1 \\ \text{Backward} & \rightarrow \frac{n(n+1)}{2} \\ \text{Forward} & \rightarrow \frac{n(n+1)}{2} \end{array} \right\}$$

when to use backward and forward

→ 563 features

i) you want 50 best features

⇒ backward

Ex:- 100 features

→ 90 best ⇒ backward

→ 10 best ⇒ forward

ii) you want 100 best
⇒ Forward

Disadvantage:-

- misses feature interaction
- Once added, a feature cannot be removed

Efficient but greedy → No Guaranteed Optimal solution.

Advantage of Wrapper Methods:-

- Accuracy :- provides best performing feature subset for ML.
- Interaction of Features :-
 - while Filter methods consider each feature independently
 - wrapper methods can find group of features that together improve the performance of model.

Disadvantages:-

- Computational Complexity :- time consuming for datasets with a large number of features.
- Risk of overfitting :-
 - They optimize the feature subset to maximize the performance of a specific ML model
 - They might select a feature subset that performs well on training data but not as well on unseen data, leading to overfitting.
- Model Specific :-
 - The selected feature subset maximize the performance of specific ML model,
 - Therefore, this subset might not perform well with a different type of model.